

Measuring Mitosis

Student worksheet | Biology 2e, Chapter 10 | 40-55 minutes

Name: _____ Class: _____ Date: _____

1. Predict and plan

What can a snapshot of dividing cells tell us about the cell cycle?

Classify cells in two schematic root-tip fields or count prepared classroom slides. Use stage frequencies to estimate time in each stage under an explicitly assumed cycle duration.

Prediction and scientific reasoning:

Variables and controls, or sampling plan:

Keep consistent: Use the same scoring definitions and count every cell in each selected field. Group prometaphase with prophase; count telophase/cytokinesis as one event until daughter cells are separated.

Select your route

Part	Virtual / Classroom	Equipment or conditions
Field A		
Field B		

Use the same questions for either route. Always label the source of each dataset.

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2. Collect observations

Record a tally for two non-overlapping fields. The virtual fields contain 20 cells each; your classroom counts may differ. Group prometaphase with prophase.

Stage	Field A	Field B	Total	Percent	Est. hours
Interphase					
Prophase					
Metaphase					
Anaphase					
Telophase					
Total				100%	

Field A data source: _____ Field B data source: _____

Assumed cycle duration: _____ hours. State why this is an assumption:

Recognition notes

Describe chromosome features used to classify two cells; record exclusions and uncertain cases.

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3. Explain the evidence

Estimated stage time = stage count / total count x assumed cycle duration. This assumes representative sampling of an asynchronous, steadily cycling population with comparable cycle lengths. Non-cycling cells, sampling bias, and differential cell loss can invalidate the estimate.

1. Which stage is most common? Give its count, percentage, and estimated duration, and distinguish observation from inference.

2. Describe the chromosome evidence you used to distinguish metaphase from anaphase. Why does interphase count even though it is not mitosis?

3. What assumptions are required to infer time from stage frequency? Explain why 24 hours is not a measured onion cell-cycle duration here.

4. Compare the two fields and identify one sampling limitation. Explain how you would improve the count.

Before sharing: check units, source labels, calculations, and evidence. The app saves on this device only; download or print your report.